

Project 1: Double-descent experiment

00103335: Deep Learning and Reinforcement Learning — Fall 2026

Method (a), (c). For each $s \in \{5, 2, 1\}$, set $\beta = \sqrt{s} e_1$, $n = 200$, and use

$$p \in \{20, 40, 60, 80, 100, 120, 140, 160, 180, 199, 200, 220, 240, 260, 300, 400, 600\}.$$

For every (s, p) , perform 20 independent repetitions. In each repetition, draw fresh, mutually independent $X \in \mathbb{R}^{200 \times p}$, $\varepsilon \in \mathbb{R}^{200}$, and $X_{\text{test}} \in \mathbb{R}^{2000 \times p}$ with independent standard Gaussian entries. Form $y = X\beta + \varepsilon$, compute the minimum-norm solution $\hat{\beta} = X^+ y$ using `numpy.linalg.lstsq`, and record

$$R = \frac{1}{2000} \|X_{\text{test}}(\hat{\beta} - \beta)\|_2^2.$$

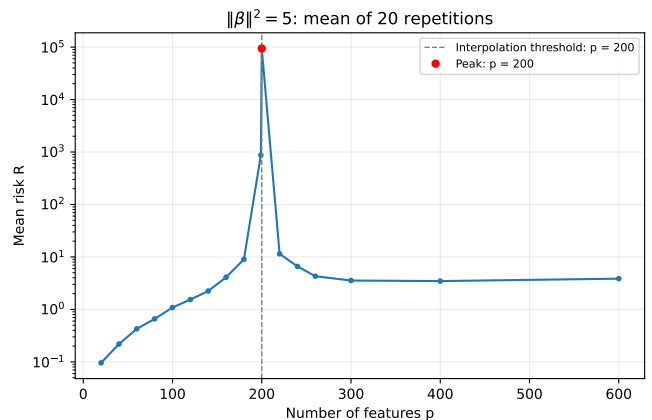
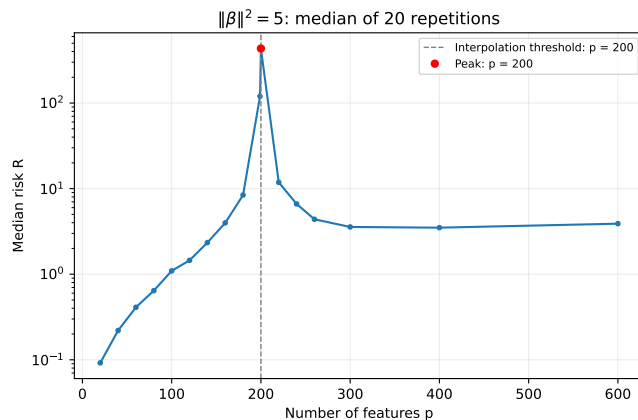
The test target is noiseless, as specified. The random seed is 20260914; `results/risks.csv` contains all 1,020 risks.

Peaks (b), (c). Each plot below uses the 20 recorded risks at each p . The vertical axis is logarithmic; medians and arithmetic sample means are computed on the original risk scale. Their observed maxima are:

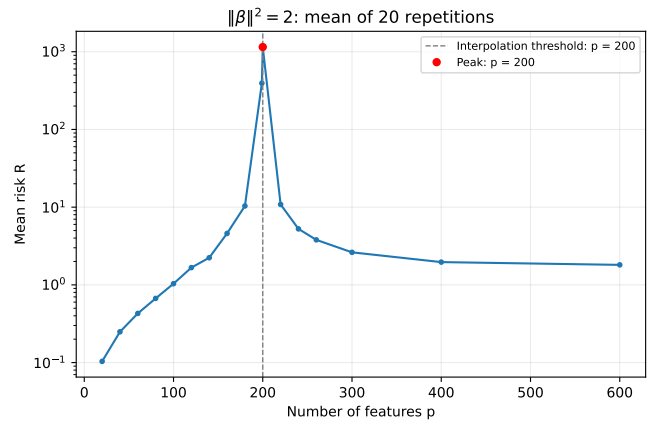
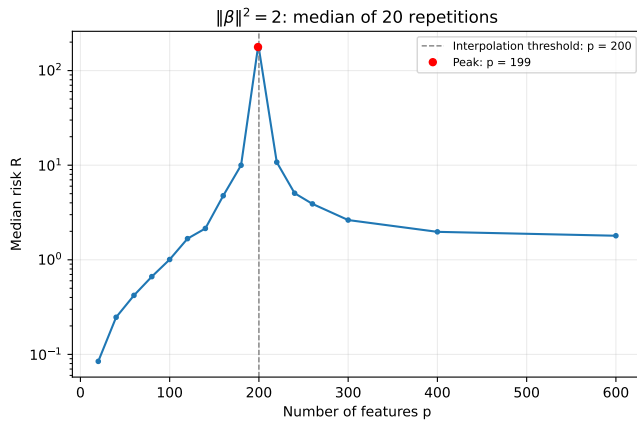
β	Median peak p	Median risk	Mean peak p	Mean risk
$\sqrt{5} e_1$	200	433.826	200	93888.507
$\sqrt{2} e_1$	199	177.463	200	1149.604
e_1	200	448.124	200	5586.063

Risk rises sharply near the interpolation threshold $p = n = 200$ and falls on the overparameterized side. For $\beta = \sqrt{2} e_1$, the observed median peaks at 199, slightly above the median 170.975 at 200. These are finite-sample peaks from 20 repetitions. The much larger mean than median near the threshold reflects extreme risks in some repetitions.

(b) $\beta = \sqrt{5} e_1$.



(c) $\beta = \sqrt{2}e_1$.



(c) $\beta = e_1$.

